ARTEMIS HACKATHON 2026

TITLE PAGE

- **Problem Statement No –1**
- **Problem Statement Title-**
- **PS Category- Software/Hardware**
- **Team Name – FALCON 0.0**

PROBLEM STATEMENT

- Farmers are losing a significant portion of their crops because plant diseases are detected too late or diagnosed incorrectly. Existing AI tools provide predictions, but they don't explain the reasoning, work poorly in rural conditions, and fail to support real farming decisions.
- The real problem isn't just disease detection — it's lack of trust, accessibility, and localized decision support.
- LeafLens AI changes this. We combine explainable crop diagnosis, farmer feedback learning, trust scores, local language support, offline-first rural design, and hyper-local weather and mandi insights — all in one intelligent, farmer-first system.

"Because Farmers Deserve to Know Why."

PROPOSED SOLUTION

How it works:

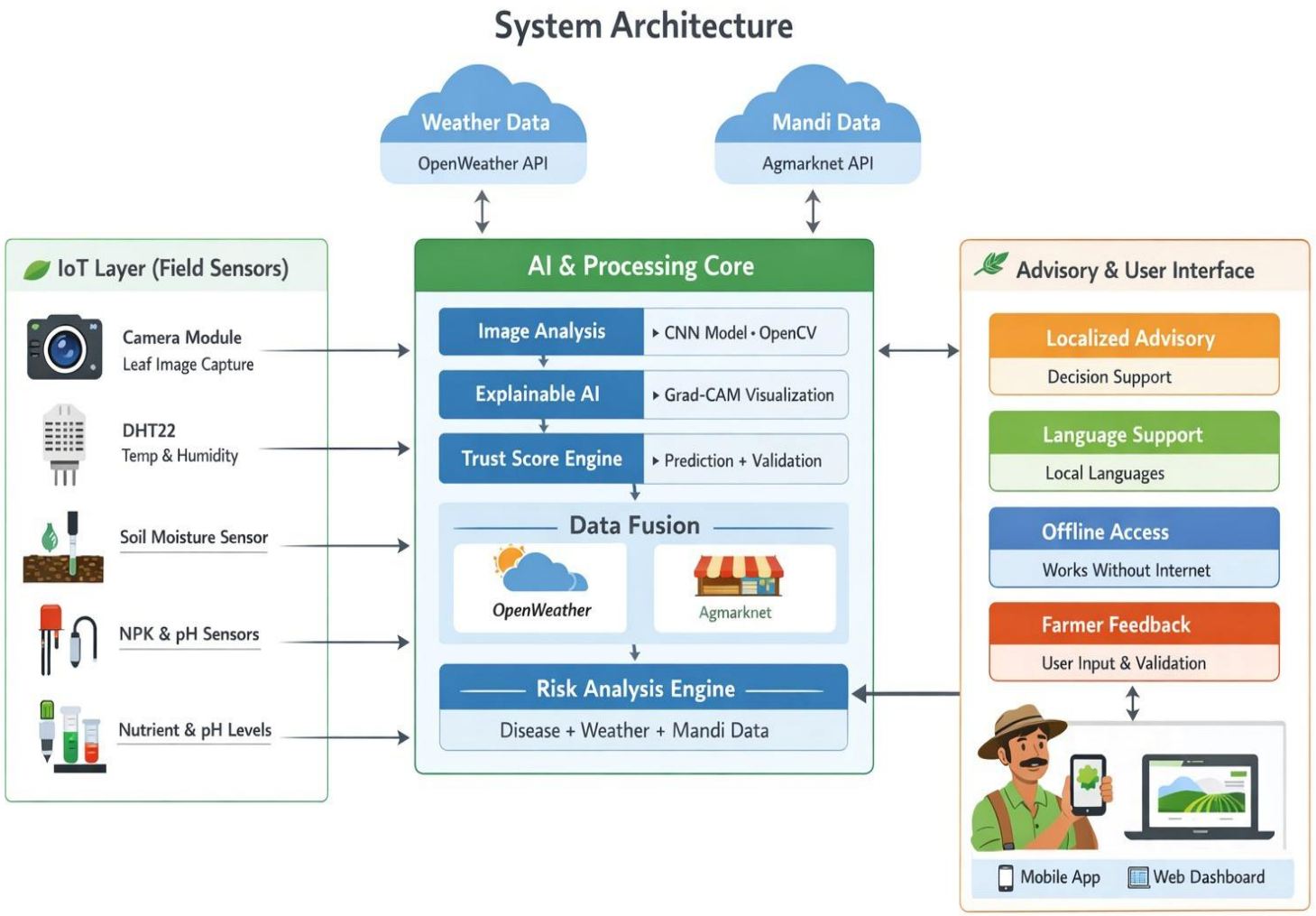
- Farmer captures leaf image
- AI detects disease
- System explains the diagnosis with confidence score
- Farmer validates feedback
- App provides weather alerts & mandi price insights
- Works in low or no internet areas

Main Features:

- Image-Based Disease Detection
- Explainable AI Diagnosis
- Trust Score System
- Farmer Feedback Learning
- Local Language Support
- Offline-First Design
- Hyper-Local Weather Alerts
- Mandi Price Decision Support

TECHNICAL APPROACH

- **Programming Languages:**
- Python – For AI model development and backend processing.
- C/C++ (Arduino IDE) – For ESP32 IoT programming.
- JavaScript – For frontend interaction and dashboard.
- **AI & ML Frameworks:**
- TensorFlow / Keras – For crop disease image classification model.
- OpenCV – For image preprocessing and enhancement.
- Grad-CAM – For Explainable AI visualization.
- **Web & Backend Frameworks:**
- Flask / FastAPI – Backend API development.
- HTML, CSS, Bootstrap – User interface design.
- REST APIs – Weather and mandi data integration
- **Database & Cloud:**
- Firebase / MongoDB – Real-time data storage.
- AWS / Render – Cloud deployment.



FEASIBILITY AND VIABILITY

Feasibility Analysis:

- Smartphone penetration in rural areas is rapidly increasing.
- Pre-trained AI models for plant disease detection already exist and can be optimized.
- Offline-first mobile architecture is technically achievable.
- Local language integration is scalable using NLP tools.
- Weather and mandi data APIs are available for integration.
- Technically achievable with current AI & mobile technology stack.

Potential Challenges & Risks:

- Limited or poor-quality crop image datasets
- Low farmer trust in AI recommendations
- Internet connectivity issues in rural areas
- Model bias due to regional crop variations
- Adoption resistance due to digital literacy gaps

Strategies to Overcome Challenges

Build region-specific datasets through farmer feedback loops
Use Explainable AI to increase transparency and trust
Implement offline-first + lightweight model deployment
Train models with diverse crop and climate data

CHALLENGES & RISKS

- **Limited Labeled Dataset Availability**

Insufficient high-quality, region-specific crop disease images may affect model accuracy.

- **Rural Connectivity Issues**

Poor or unstable internet access can limit real-time updates and API integration.

- **Farmer Trust Deficit in AI**

Lack of explainability in AI systems may reduce adoption and confidence.

- **Weather Prediction Uncertainty**

Inaccurate or fluctuating weather forecasts can impact advisory reliability.

- **Sensor Calibration & Maintenance**

IoT hardware may face wear, environmental damage, or inaccurate readings over time.

- **Digital & Language Barriers**

Low digital literacy and absence of regional language support can hinder usability.

- **Data Privacy & Ownership Concerns**

Farmers may worry about misuse of agricultural data or lack of control over information.

- **Scalability & Deployment Cost**

Large-scale implementation across multiple regions may require infrastructure investment.

IMPACT AND BENEFITS

Potential Impact :

- Improved Disease Detection Accuracy
- Data-Driven Decision Making
- Increased Farmer Confidence
- Empowerment of Small & Marginal Farmers
- Enhanced Support for Agri-Extension Officers

Benefits of the Solution:

- **Social Benefits:**

Promotes digital inclusion in rural communities through local language and offline-first design.Improves farmer awareness and technological adoption in agriculture.

- **Economic Benefits:**

Reduces crop loss (40–60%) through early detection and risk-aware advisory.Helps farmers choose the right time and market to sell crops for better profits.Minimizes dependency on intermediaries, increasing direct income.

- **Environmental Benefits:**

Encourages precise pesticide usage, reducing chemical overuse.Supports climate-resilient farming through hyper-local weather alerts.Promotes sustainable soil management using IoT-based monitoring.

TARGET USERS

Target Users

Small & Marginal Farmers

Farmers with **limited resources** who need **affordable, easy-to-use,** and reliable decision-support tools.

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Rural Agricultural Communities

Villages facing connectivity challenges and requiring **offline-first**, local-language

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Agri-Extension Officers

Field officers who support multiple farmers and need accurate, explainable AI-based advisory tools.

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Farmer Producer Organizations (FPOs)

Groups that can use **integrated disease**, weather, and **mandi insights** for collective decision-making.

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Government & Agricultural Departments

Institutions aiming to improve crop productivity, reduce losses, and promote **smart farming initiatives**.

FUTURE SCOPE

- Expansion to Multi-Crop & Multi-Region Support
- Advanced AI & Predictive Analytics
- Works in low or no internet areas
- Mobile App with Voice Assistance
- Satellite & Drone Integration
- Government & FPO Integration
- Agri-Finance & Insurance Advisory
- Sustainable Farming Recommendations
- AI-Powered Marketplace Integration

BUSINESS MODEL / PROFIT STRATEGY

Business Model:

- **Freemium Model for Farmers:**Basic disease detection and advisory features provided free to ensure accessibility.
- **Premium Subscription Plan:**Advanced features like predictive analytics, detailed mandi forecasting, and IoT integration offered under affordable subscription plans.
- **B2B Partnerships:**Collaborate with Farmer Producer Organizations (FPOs), agri-startups, NGOs, and cooperatives for bulk deployment.
- **Government & Institutional Collaboration:**Partner with agriculture departments and rural development schemes for large-scale implementation.

Revenue Streams:

- **Farmer Premium Subscriptions:**Low-cost monthly/seasonal plans for advanced advisory services.
- **FPO & Cooperative Licensing:**Annual licensing model for group-based usage.
- **IoT Hardware Sales / Rental:**Revenue from sensor kits (ESP32 + Soil sensors) or subscription-based hardware leasing.
- **API Integration Services:**Paid API access for agri-tech companies and digital platforms.
- **Government Grants & Agri Innovation Funding:**Support through smart agriculture and Digital India initiatives.

RESEARCH AND REFERENCES

- **AI-Based Crop Disease Detection**
- Too et al. (2020), Computers & Electronics in Agriculture – CNN models for plant disease classification.
- Saleem et al. (2022), Plants (MDPI) – Deep learning advancements in crop disease detection.
- **Explainable AI (XAI)**
- Arrieta et al. (2020), Information Fusion – Foundations of Explainable AI.
- Grad-CAM (2020–2023 applications) – Visual explanation technique for CNN models.
- **Smart Agriculture & IoT**
- Liakos et al. (2021), Sensors Journal – ML & IoT applications in agriculture.
- Wolfert et al. (2020–2022), Agricultural Systems – Big data in smart farming.
- **Climate & Indian Context**
- IPCC (2022) – Climate change impact on crop diseases.
- Digital Agriculture Mission, Govt. of India (2021–2025) – AI & IoT in Indian agriculture.
- Agmarknet (2020–2024) – Official mandi price database.

IMPORTANT INSTRUCTIONS

SMART INDIA HACKATHON 2024

Please ensure below pointers are met while submitting the Idea PPT:

1. **Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures**
2. **Keep your explanation precise and easy to understand**
3. **You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).**

Note - You can delete this slide (Important Pointers) when you give the seminar.